

Supplementary Information for
“Low power and overrepresentation of statistically significant findings in the
environmental sciences”

Teshome K. Deressa, David I. Stern, Jaco Vangronsveld, Jan Minx, Sebastien Lizin,
Robert Malina, Stephan B. Bruns

May 1, 2026

Outline:

Appendix A. Results for power using different values of the genuine effect

Appendix B. Heterogeneity by subfield

Appendix C. Extent of selective reporting for the subfields

Appendix D. Sensitivity analyses for regression results along diagnostic plots

Appendix E. Types of effect sizes used in the included meta-analyses

Appendix A.

Table S1: Statistical power in environmental sciences. Half of the full meta-average is used to approximate the genuine effect. Moreover, primary estimates that are statistically non-significant at the 5% level are excluded (Nord et al. 2017; Yang et al. 2023).

Subfield	No. of meta-analyses	No. of primary estimates	Median of medians	Median	Mean	Q25	Q75	SAPE
All meta-analyses	660	33946	0.11	0.27	0.44	0.09	0.93	0.29
Ecology	156	10606	0.10	0.29	0.45	0.08	0.94	0.30
Environmental Chemistry	62	3713	0.11	0.14	0.33	0.07	0.51	0.18
Environmental Engineering	40	7105	0.32	0.24	0.40	0.14	0.66	0.20
Health, Toxicology and Mutagenesis	250	4897	0.12	0.97	0.63	0.13	1.00	0.54
Management, Monitoring, Policy and Law	17	372	0.06	0.09	0.25	0.05	0.39	0.08
Nature and Landscape Conservation	94	6798	0.07	0.22	0.41	0.07	0.83	0.26
Water Science and Technology	41	455	0.22	0.50	0.54	0.11	1.00	0.42

Notes: Median of medians denotes the median of the medians of statistical power per meta-analysis. Meanwhile, the columns Median, Mean, Q25, Q75, and SAPE (Share of Adequately Powered Estimates) are directly based on the respective distribution of primary estimates.

Table S2: Statistical power in environmental sciences. Half of the full meta-average is used to approximate the genuine effect. Additionally, meta-analyses whose estimated pooled effects are statistically non-significant at the 5% level are excluded (Lamberink et al. 2018).

Subfield	No. of meta-analyses	No. of primary estimates	Median of medians	Median	Mean	Q25	Q75	SAPE
All meta-analyses	213	39168	0.25	0.25	0.42	0.11	0.82	0.25
Ecology	55	10854	0.45	0.34	0.47	0.12	0.92	0.29
Environmental Chemistry	26	5340	0.18	0.10	0.27	0.07	0.30	0.13
Environmental Engineering	28	10943	0.32	0.22	0.36	0.13	0.52	0.15
Health, Toxicology and Mutagenesis	59	4539	0.18	0.94	0.61	0.11	1.00	0.53
Management, Monitoring, Policy and Law	3	138	0.24	0.38	0.42	0.15	0.64	0.17
Nature and Landscape Conservation	29	7029	0.20	0.26	0.43	0.10	0.86	0.27
Water Science and Technology	13	325	0.96	0.83	0.66	0.26	1.00	0.53

Notes: Median of medians denotes the median of the medians of statistical power per meta-analysis. Meanwhile, the columns Median, Mean, Q25, Q75, and SAPE (Share of Adequately Powered Estimates) are directly based on the respective distribution of primary estimates.

Table S3: Share of subfields with median power $\geq 80\%$ and share of meta-analyses with indication for small-study effect

Subfield	Share of subfields with median power larger 80% (in %) ^a	Share of meta-analyses with small-study effects (in %) ^b
Ecology	34.9	36.4
Environmental Chemistry	6.00	41.5
Environmental Engineering	18.1	50.0
Health, Toxicology and Mutagenesis	22.9	25.4
Management, Monitoring, Policy and Law	0.00	5.3
Nature and Landscape Conservation	6.00	37.8
Water Science and Technology	12.1	19.5
All meta-analyses	11.8	31.7

Notes: ^a 83 out of 704 meta-analyses have a median power of primary estimates larger than 80%. The share of subfields in these meta-analyses is reported. ^b The share of meta-analyses with an indication for small-study effects is reported. We use a multilevel version of Egger's regression to test for small-study effects.

Subfield	M	Median	Mean	Q25	Q75	Heterogeneity
Ecology	162	88.91	79.14	71.47	98.54	
Environmental Chemistry	65	91.78	78.88	67.2	98.96	
Environmental Engineering	40	99.53	95.85	99.34	99.85	
Health, Toxicology and Mutagenesis	279	69.3	58.99	23.96	91.2	
Management, Monitoring, Policy and Law	19	82.26	75.25	75.66	91.54	
Nature and Landscape Conservation	98	82.43	71.23	59.41	96.68	
Water Science and Technology	41	94.09	81.23	80.52	96.12	

Figure S1: Distribution of heterogeneity (I^2 in percentage) in the meta-analyses by subfield.

Appendix C.

Table S4: Differences in relative frequencies and extent of selective reporting for ‘Ecology’, ‘Environmental Chemistry’, and ‘Environmental Engineering’ subfields.

<i>p</i> -value interval	Ecology		Env. Chemistry		Env. Engineering	
	Difference	95% CI	Difference	95% CI	Difference	95% CI
0.9 < <i>p</i>	-0.009	[-0.019, -0.001]	-0.006	[-0.012, 0.001]	-0.007	[-0.012, -0.001]
0.8 < <i>p</i> < 0.9	-0.020	[-0.029, -0.012]	-0.024	[-0.030, -0.017]	-0.007	[-0.012, -0.001]
0.7 < <i>p</i> < 0.8	-0.022	[-0.031, -0.014]	-0.020	[-0.026, -0.014]	-0.013	[-0.018, -0.007]
0.6 < <i>p</i> < 0.7	-0.026	[-0.035, -0.018]	-0.018	[-0.023, -0.011]	-0.023	[-0.027, -0.017]
0.5 < <i>p</i> < 0.6	-0.022	[-0.031, -0.014]	-0.018	[-0.024, -0.012]	-0.013	[-0.017, -0.007]
0.4 < <i>p</i> < 0.5	-0.020	[-0.029, -0.012]	-0.026	[-0.031, -0.019]	-0.028	[-0.032, -0.023]
0.3 < <i>p</i> < 0.4	-0.025	[-0.032, -0.017]	-0.029	[-0.034, -0.023]	-0.033	[-0.037, -0.028]
0.2 < <i>p</i> < 0.3	-0.027	[-0.034, -0.020]	-0.033	[-0.037, -0.027]	-0.039	[-0.041, -0.035]
0.1 < <i>p</i> < 0.2	-0.022	[-0.028, -0.017]	-0.032	[-0.035, -0.027]	-0.045	[-0.048, -0.040]
0.05 < <i>p</i> < 0.1	-0.010	[-0.015, -0.005]	-0.014	[-0.017, -0.010]	-0.024	[-0.027, -0.018]
0.01 < <i>p</i> < 0.05	0.004	[-0.007, 0.017]	0.009	[0.001, 0.018]	-0.032	[-0.040, -0.023]
0.001 < <i>p</i> < 0.01	0.017	[0.005, 0.031]	0.027	[0.017, 0.037]	0.006	[-0.001, 0.015]
<i>p</i> < 0.001	0.184	[0.128, 0.238]	0.184	[0.146, 0.214]	0.258	[0.221, 0.280]
$ESR_{0.05}^{all}$	0.204	[0.130, 0.272]	0.219	[0.164, 0.262]	0.232	[0.189, 0.262]
$ESR_{0.05}^{sig.}$	0.385	[0.296, 0.488]	0.502	[0.432, 0.569]	0.408	[0.345, 0.465]
No. of meta-analyses	162		65		40	
No. of tests	20003		8490		12518	

Notes: Differences between reported *p*-values and counterfactually expected *p*-values per *p*-value bins as a share of all *p*-values are reported. Negative signs indicate that *p*-values are missing in the respective bin compared to what is expected in the absence of any biases. $ESR_{0.05}^{all}$ represents the estimated extent of selective reporting as a share of all *p*-values, whereas $ESR_{0.05}^{sig.}$ represents the estimated extent of selective reporting as a share of only significant *p*-values. 95% confidence intervals are based on bootstrapping clustered at the level of meta-studies.

Table S5: Differences in relative frequencies and extent of selective reporting for ‘Health, Toxicology and Mutagenesis’, ‘Nature and Landscape Conservation’, and ‘Management, Monitoring, Policy and Law’ subfields.

<i>p</i> -value interval	Health, Tox. & Mut.		Mgmt, Mon. Pol. & Law		Nat. & Land. Cons.	
	Difference	95% CI	Difference	95% CI	Difference	95% CI
0.9 < <i>p</i>	0.033	[0.018, 0.041]	-0.012	[-0.028, 0.005]	-0.021	[-0.034, -0.009]
0.8 < <i>p</i> < 0.9	-0.015	[-0.030, -0.007]	-0.037	[-0.052, -0.020]	-0.025	[-0.037, -0.013]
0.7 < <i>p</i> < 0.8	-0.019	[-0.033, -0.010]	-0.027	[-0.043, -0.011]	-0.017	[-0.029, -0.005]
0.6 < <i>p</i> < 0.7	-0.020	[-0.035, -0.012]	-0.040	[-0.055, -0.024]	-0.023	[-0.035, -0.011]
0.5 < <i>p</i> < 0.6	-0.019	[-0.035, -0.011]	-0.022	[-0.036, -0.007]	-0.018	[-0.030, -0.006]
0.4 < <i>p</i> < 0.5	-0.024	[-0.040, -0.015]	-0.044	[-0.057, -0.031]	-0.022	[-0.034, -0.011]
0.3 < <i>p</i> < 0.4	-0.024	[-0.040, -0.014]	-0.028	[-0.039, -0.017]	-0.028	[-0.039, -0.018]
0.2 < <i>p</i> < 0.3	-0.027	[-0.045, -0.016]	-0.049	[-0.055, -0.041]	-0.030	[-0.040, -0.020]
0.1 < <i>p</i> < 0.2	-0.019	[-0.039, -0.007]	-0.036	[-0.041, -0.032]	-0.032	[-0.040, -0.025]
0.05 < <i>p</i> < 0.1	-0.007	[-0.020, 0.001]	0.009	[-0.001, 0.020]	-0.014	[-0.018, -0.011]
0.01 < <i>p</i> < 0.05	0.027	[0.013, 0.038]	0.010	[-0.018, 0.038]	0.001	[-0.009, 0.012]
0.001 < <i>p</i> < 0.01	0.024	[0.016, 0.030]	0.021	[-0.006, 0.048]	0.016	[0.001, 0.033]
<i>p</i> < 0.001	0.091	[-0.017, 0.272]	0.253	[0.203, 0.290]	0.214	[0.139, 0.296]
$ESR_{0.05}^{all}$	0.141	[0.050, 0.300]	0.285	[0.184, 0.377]	0.231	[0.135, 0.332]
$ESR_{0.05}^{sig.}$	0.236	[0.146, 0.396]	0.626	[0.450, 0.791]	0.445	[0.320, 0.606]
No. of meta-analyses	279		19		98	
No. of tests	8220		818		13115	

Notes: Differences between reported *p*-values and counterfactually expected *p*-values per *p*-value bins as a share of all *p*-values are reported. Negative signs indicate that *p*-values are missing in the respective bin compared to what is expected in the absence of any biases. $ESR_{0.05}^{all}$ represents the estimated extent of selective reporting as a share of all *p*-values, whereas $ESR_{0.05}^{sig.}$ represents the estimated extent of selective reporting as a share of only significant *p*-values. 95% confidence intervals are based on bootstrapping clustered at the level of meta-studies.

Table S6: Differences in relative frequencies and extent of selective reporting for the ‘Water Science and Technology’ subfield.

<i>p</i> -value interval	Difference	95% CI
0.9 < <i>p</i>	-0.023	[-0.034, -0.009]
0.8 < <i>p</i> < 0.9	-0.009	[-0.020, 0.004]
0.7 < <i>p</i> < 0.8	-0.028	[-0.038, -0.014]
0.6 < <i>p</i> < 0.7	-0.012	[-0.023, 0.000]
0.5 < <i>p</i> < 0.6	-0.015	[-0.025, -0.002]
0.4 < <i>p</i> < 0.5	-0.020	[-0.031, -0.007]
0.3 < <i>p</i> < 0.4	-0.025	[-0.036, -0.012]
0.2 < <i>p</i> < 0.3	-0.018	[-0.029, -0.006]
0.1 < <i>p</i> < 0.2	-0.028	[-0.039, -0.017]
0.05 < <i>p</i> < 0.1	0.004	[-0.004, 0.012]
0.01 < <i>p</i> < 0.05	0.042	[0.024, 0.056]
0.001 < <i>p</i> < 0.01	-0.005	[-0.026, 0.013]
<i>p</i> < 0.001	0.137	[0.031, 0.233]
$ESR_{0.05}^{all}$	0.174	[0.058, 0.273]
$ESR_{0.05}^{sig.}$	0.298	[0.196, 0.424]
No. of meta-analyses	41	
No. of tests	792	

Notes: Differences between reported *p*-values and counterfactually expected *p*-values per *p*-value bins as a share of all *p*-values are reported. Negative signs indicate that *p*-values are missing in the respective bin compared to what is expected in the absence of any biases. $ESR_{0.05}^{all}$ represents the estimated extent of selective reporting as a share of all *p*-values, whereas $ESR_{0.05}^{sig.}$ represents the estimated extent of selective reporting as a share of only significant *p*-values. 95% confidence intervals are based on bootstrapping clustered at the level of meta-studies.

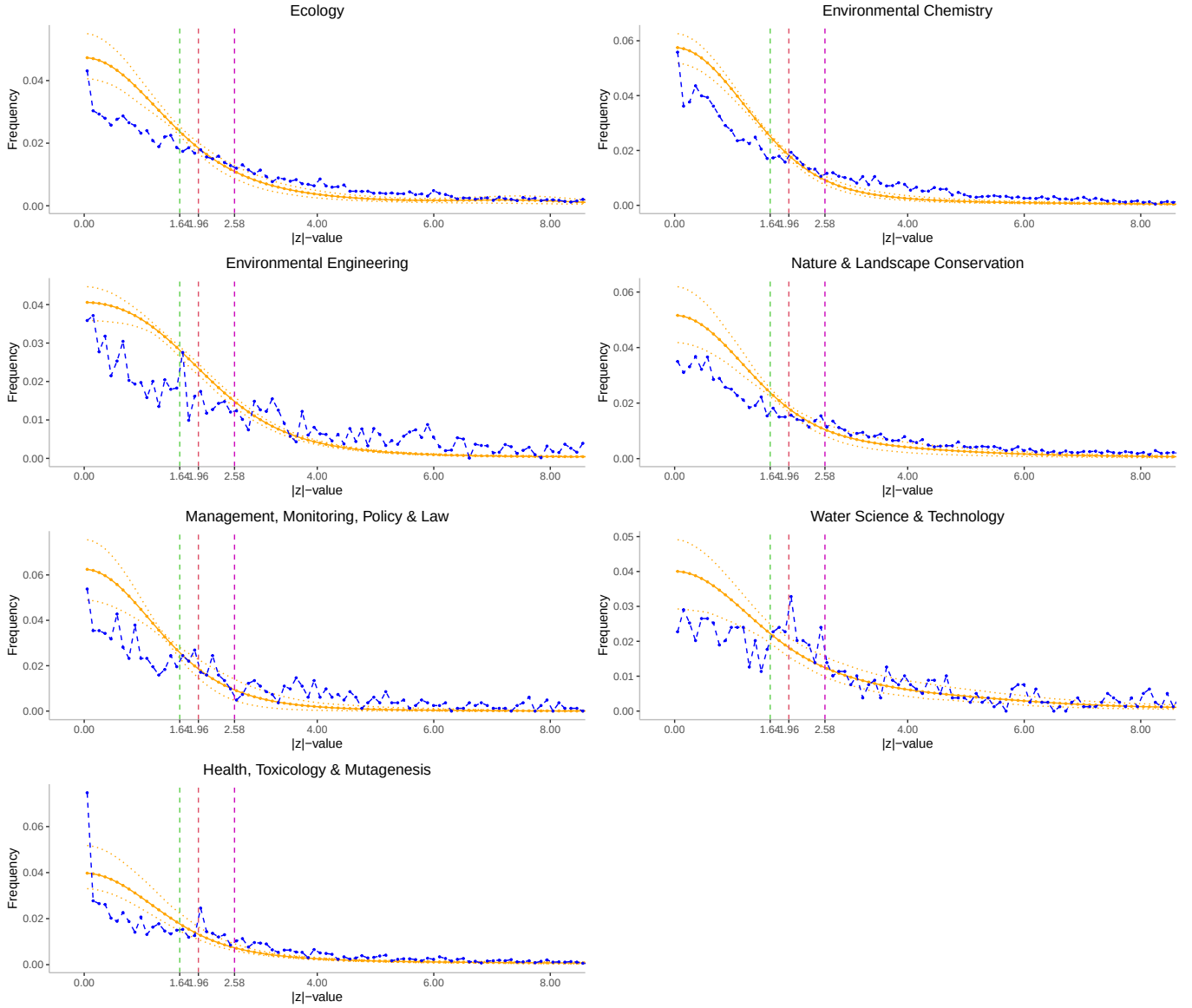


Figure S2: Distributions of factual and counterfactual z -values broken down by subfields. Notes: Factual distributions are given by the blue dashed lines and counterfactual distributions are given by the orange solid lines with dashed 0.95 confidence intervals based on bootstrapping clustered by meta-studies. The dots are placed at the centre of each interval using a grid of 0.1025. This interval size ensures that the critical value of 1.64 represents an interval border. The dashed vertical lines represent the thresholds of statistical significance for the 0.1 ($z = 1.64$), 0.05 ($z = 1.96$), and 0.01 ($z = 2.58$) levels.

Table S7: Negative binomial regression results. A full meta-average is used to approximate genuine effect.

Variable	Model 1	Model 2
	Estimate (SE)	Estimate (SE)
Intercept	54.692* (31.608)	25.793 (28.478)
Median power	−0.027*** (0.002)	−0.026*** (0.002)
Experimental research design? (yes)	0.092 (0.074)	0.062 (0.072)
Followed reporting guidelines? (yes)	0.100 (0.067)	0.025 (0.080)
Protocol registered? (yes)	−0.108 (0.113)	−0.147 (0.123)
Log number of independent studies	−0.062 (0.042)	−0.026 (0.041)
Log journal impact factor	−0.069 (0.062)	−0.050 (0.067)
Publication year	−0.027* (0.016)	−0.013 (0.014)
Environmental Chemistry		−0.207* (0.109)
Environmental Engineering		−0.481** (0.209)
Health, Toxicology and Mutagenesis		−0.005 (0.109)
Management, Monitoring, Policy and Law		0.086 (0.073)
Nature and Landscape Conservation		−0.044 (0.050)
Water Science and Technology		−0.047 (0.224)
Effect size type	Yes	Yes
AIC	2249.5	2245.8
No. of meta-analyses	482	482

Notes: * $p < .1$; ** $p < .05$; *** $p < .01$. . For the two models, our dependent variable is $\ln ESR_{M_{0.05}}$, that is, log of the difference between the observed number of significant p -values and the expected number of significant p -values. The sample is restricted to $ESR_{M_{0.05}} \geq 0$ and the log number of significant p -values is used as an offset variable when fitting the models. We cluster standard errors at the level of meta-studies. 'Experimental' is an indicator variable denoting whether the designs of primary studies included in the meta-analysis are experimental. For the subfield, 'Ecology' is considered as a reference category. The type of effect size used in the meta-analyses is included as a control variable in both models. For all models, our dependent variable is $ESR_{M_{0.05}}^{sig}$, that is, the estimated extent of selective reporting expressed as a share of the significant p -values at the 0.05 level. We cluster standard errors at the level of meta-studies. 'Experimental' is an indicator variable denoting whether the designs of primary studies included in the meta-analysis are experimental. For the subfield, 'Ecology' is considered a reference category for research design.

Table S8: Negative binomial regression results. Half of the full meta-average is used to approximate genuine effect, and 25% of the between-study heterogeneity is considered when generating the counterfactual z -values.

Variable	Model 1	Model 2
	Estimate (SE)	Estimate (SE)
Intercept	1.558 (12.300)	−6.964 (14.269)
Median power	−0.012*** (0.001)	−0.012*** (0.001)
Experimental research design? (yes)	−0.036 (0.032)	−0.059* (0.031)
Followed reporting guidelines? (yes)	−0.014 (0.034)	0.001 (0.040)
Protocol registered? (yes)	−0.033 (0.041)	−0.064 (0.044)
Log number of independent studies	0.021 (0.015)	0.031** (0.015)
Log journal impact factor	−0.015 (0.037)	0.005 (0.036)
Publication year	−0.001 (0.006)	0.003 (0.007)
Environmental Chemistry		−0.061* (0.037)
Environmental Engineering		−0.084 (0.055)
Health, Toxicology and Mutagenesis		−0.143 (0.087)
Management, Monitoring, Policy and Law		0.115** (0.046)
Nature and Landscape Conservation		0.012 (0.038)
Water Science and Technology		−0.037 (0.096)
Effect size type	Yes	Yes
AIC	3344.5	3337.0
No. of meta-analyses	681	681

Notes: * $p < .1$; ** $p < .05$; *** $p < .01$. . For the two models, our dependent variable is $\ln ESR_{M_{0.05}}$, that is, log of the difference between the observed number of significant p -values and the expected number of significant p -values. The sample is restricted to $ESR_{M_{0.05}} \geq 0$ and the log number of significant p -values is used as an offset variable when fitting the models. We cluster standard errors at the level of meta-studies. 'Experimental' is an indicator variable denoting whether the designs of primary studies included in the meta-analysis are experimental. For the subfield, 'Ecology' is considered as a reference category. The type of effect size used in the meta-analyses is included as a control variable in both models. For all models, our dependent variable is $ESR_{M_{0.05}}^{sig}$, that is, the estimated extent of selective reporting expressed as a share of the significant p -values at the 0.05 level. We cluster standard errors at the level of meta-studies. 'Experimental' is an indicator variable denoting whether the designs of primary studies included in the meta-analysis are experimental. For the subfield, 'Ecology' is considered a reference category for research design.

Table S9: Regression results using OLS. The dependent variable winsorized at 5th and 95th percentiles.

Variable	Model 1	Model 2
	Estimate (SE)	Estimate (SE)
Intercept	3.700 (17.473)	1.611 (18.364)
Median power	−0.720*** (0.047)	−0.706*** (0.049)
Experimental research design? (yes)	0.021 (0.039)	0.012 (0.039)
Followed reporting guidelines? (yes)	0.009 (0.051)	0.008 (0.061)
Protocol registered? (yes)	−0.029 (0.064)	−0.030 (0.065)
Log number of independent studies	0.041** (0.018)	0.048*** (0.018)
Log journal impact factor	−0.051 (0.033)	−0.054 (0.038)
Publication year	−0.002 (0.009)	−0.0005 (0.009)
Environmental Chemistry		−0.001 (0.041)
Environmental Engineering		−0.056 (0.059)
Health, Toxicology and Mutagenesis		0.010 (0.080)
Management, Monitoring, Policy and Law		0.078 (0.048)
Nature and Landscape Conservation		−0.005 (0.056)
Water Science and Technology		−0.005 (0.070)
Effect size type	Yes	Yes
Adjusted R^2	0.411	0.409
No. of meta-analyses	704	704

Notes: * $p < .1$; ** $p < .05$; *** $p < .01$. For all models, our dependent variable is $ESR_{0.05}^{sig.}$, that is, the estimated extent of selective reporting expressed as a share of the significant p -values at the 0.05 level. We cluster standard errors at the level of meta-studies. 'Experimental' is an indicator variable denoting whether the designs of primary studies included in the meta-analysis are experimental. For the subfield, 'Ecology' is considered a reference category for research design.

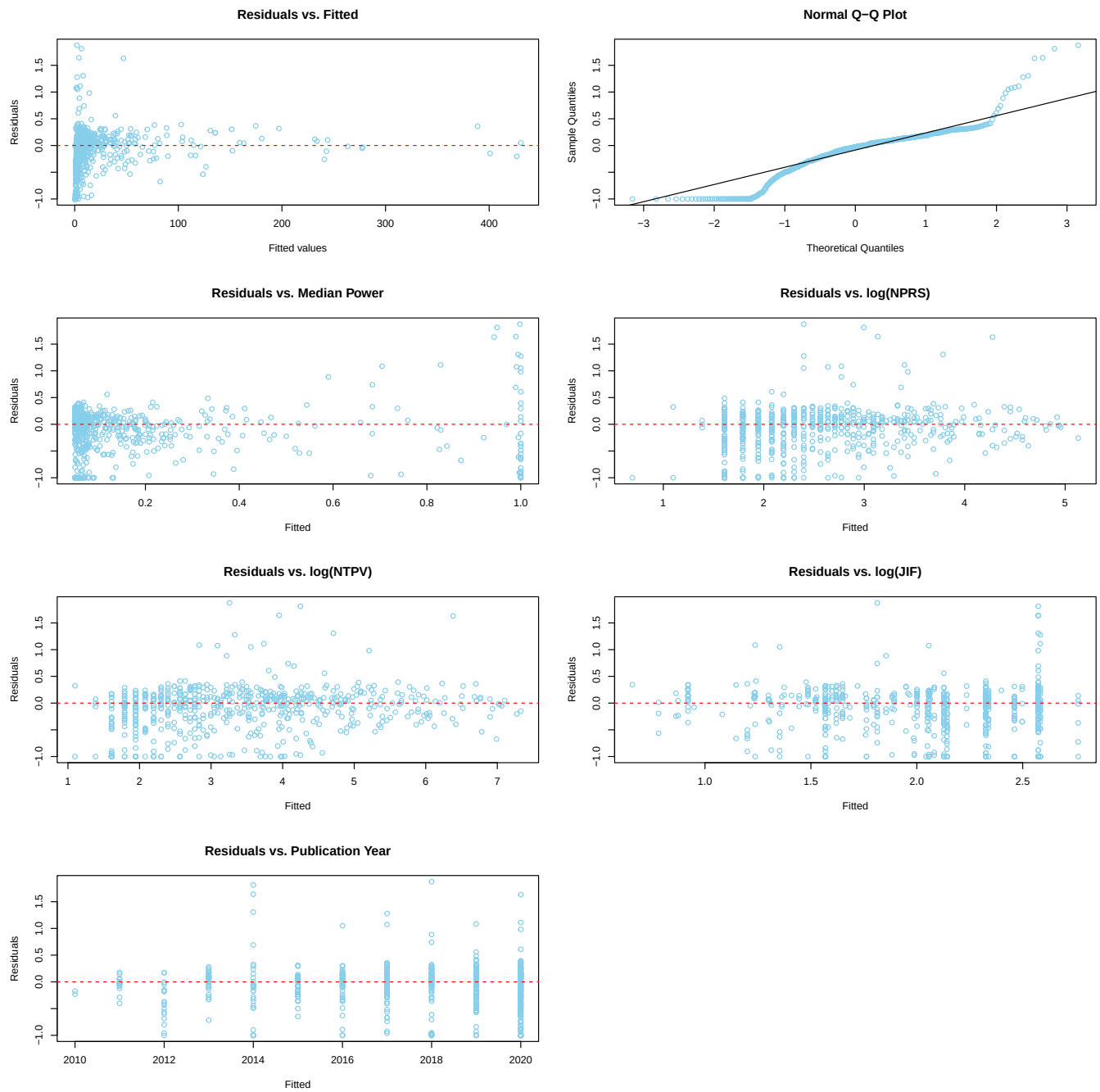


Figure S3: Plots of residuals versus continuous predictors based on Model 1 reported in the main document.

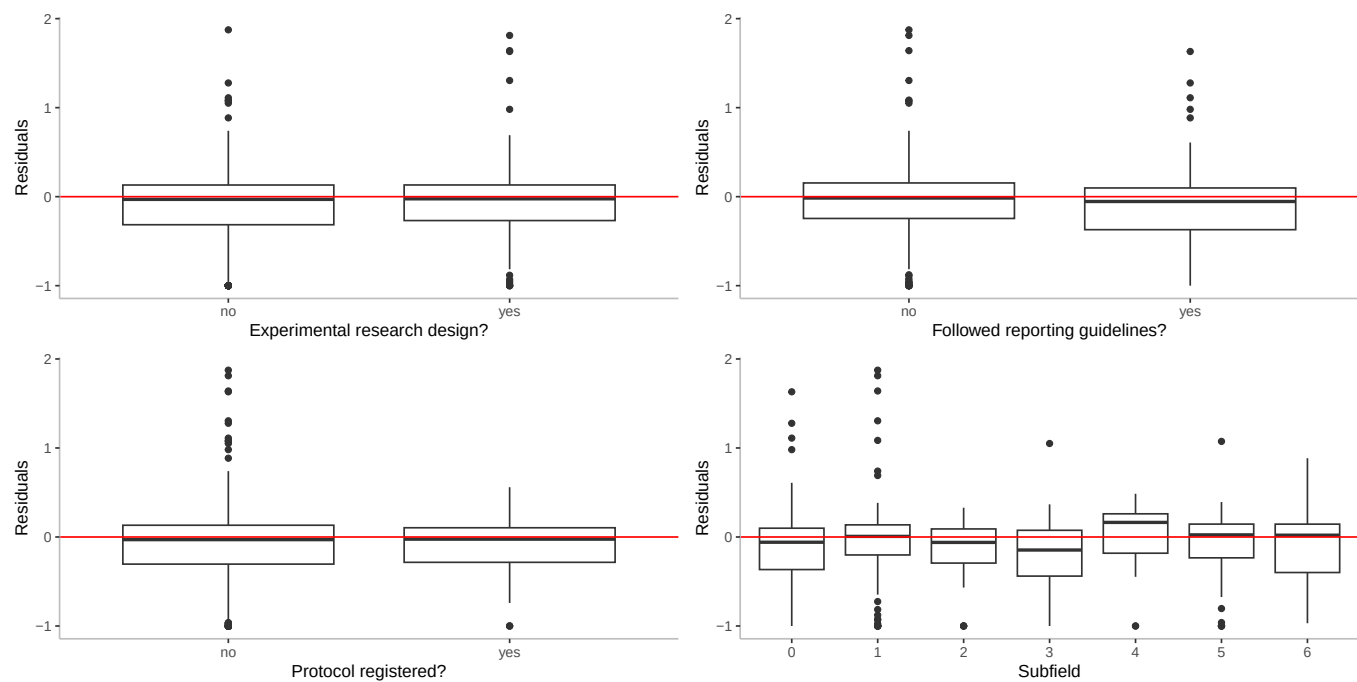


Figure S4: Plots of residuals versus categorical predictors based on Model 1 reported in the main document. 0 = Health, Toxicology and Mutagenesis, 1 = Ecology, 2 = Environmental Chemistry, 3 = Environmental Engineering, 4 = Management, Monitoring, Policy and Law, 5 = Nature and Landscape Conservation, and 6 = Water Science and Technology.

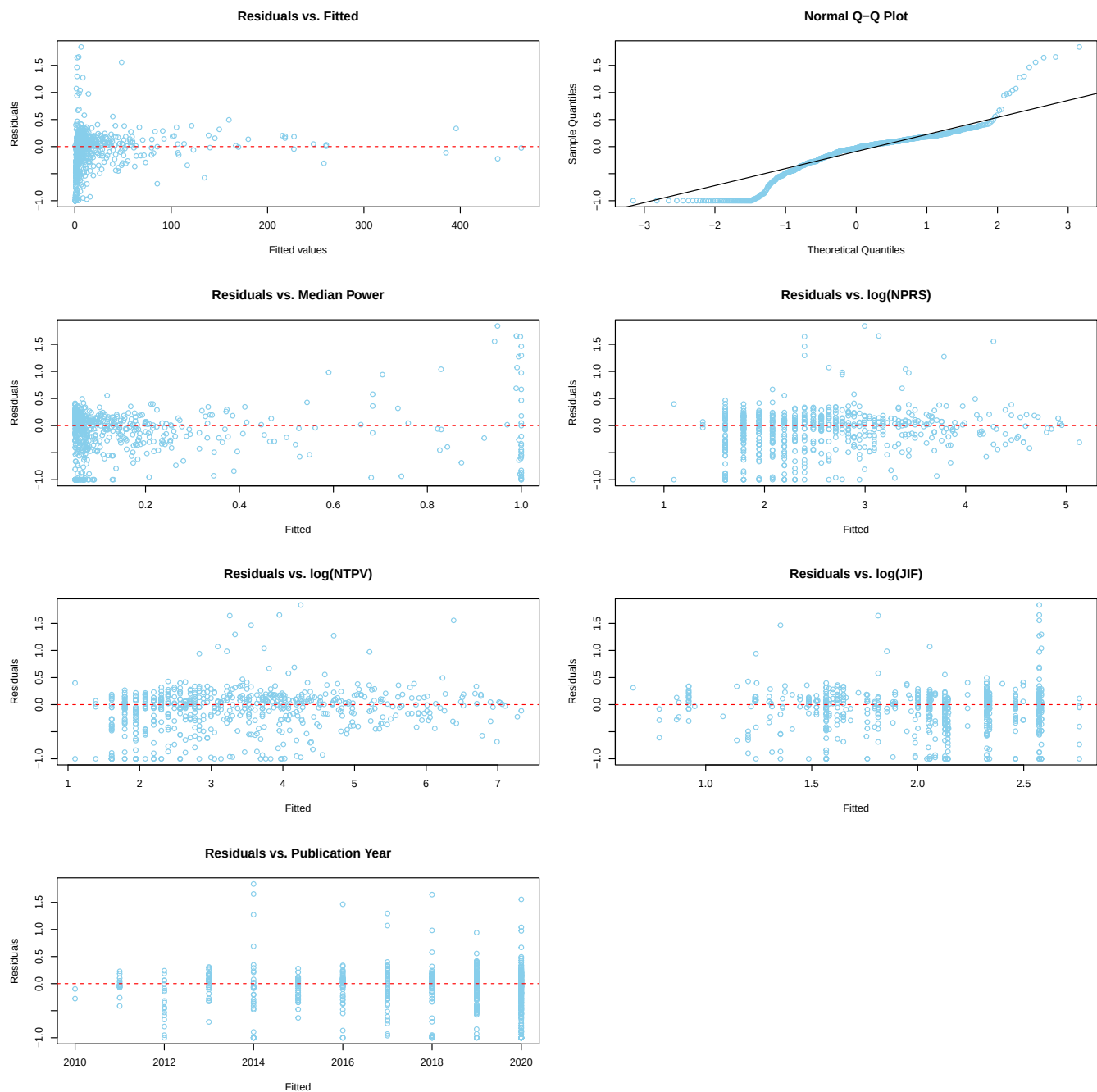


Figure S5: Plots of residuals versus continuous predictors based on Model 2 reported in the main document.

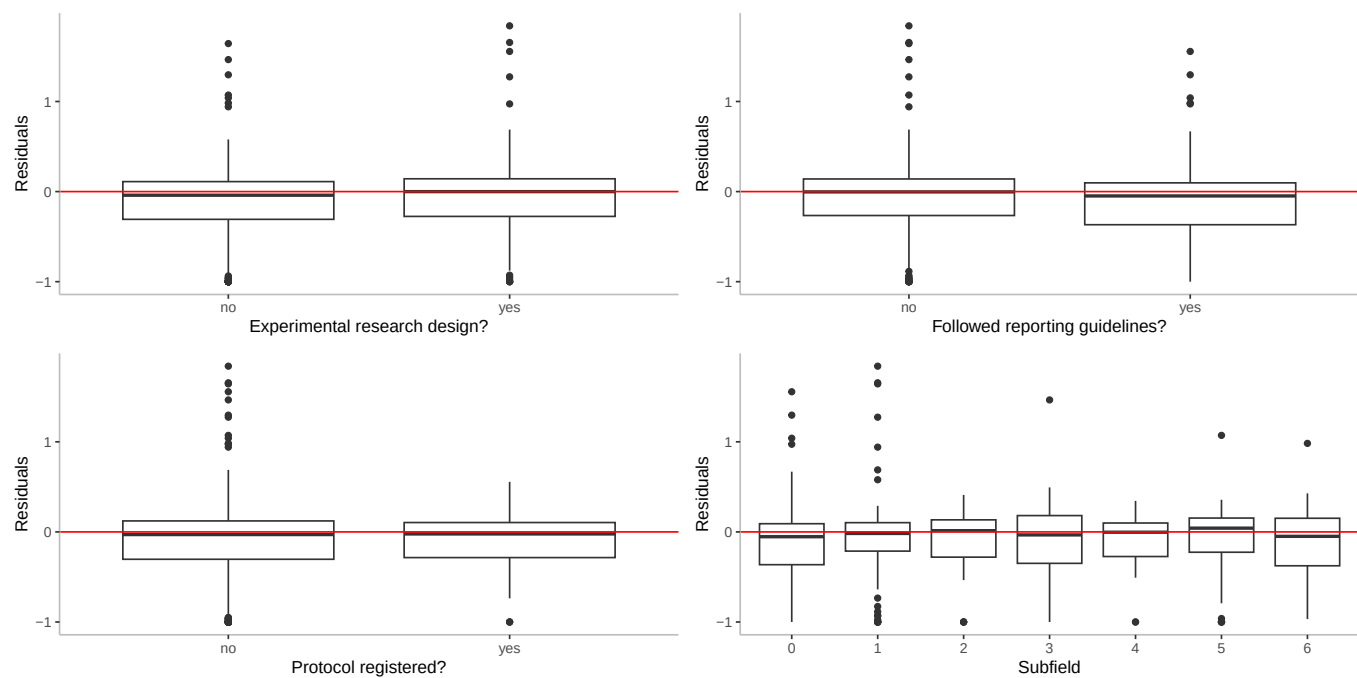


Figure S6: Plots of residuals versus categorical predictors based on Model 2 reported in the main document. 0 = Health, Toxicology and Mutagenesis, 1 = Ecology, 2 = Environmental Chemistry, 3 = Environmental Engineering, 4 = Management, Monitoring, Policy and Law, 5 = Nature and Landscape Conservation, and 6 = Water Science and Technology.

Table S10: Number of effect size used in the meta-analyses by subfields

No.	Effect size	ECO	ENC	ENE	HTM	MPL	NLC	WST
1	Cohen's d	5	9	3	2	0	2	2
2	Hedge's g	55	13	1	22	0	51	1
3	Correlation coefficient (r)	5	0	0	1	0	0	15
4	Fisher's z	4	2	0	0	0	8	1
5	Log-response ratio (lnRR)	93	32	36	23	19	31	2
6	Log-odds ratio (logOR)	0	2	0	112	0	4	17
7	Log-relative risk (logRR)	0	3	0	34	0	2	0
8	Log-hazard ratio (logHR)	0	0	0	7	0	0	0
9	Raw mean or mean difference	0	3	0	3	0	0	2
10	Percentage change	0	1	0	24	0	0	0
11	Excess risk	0	0	0	14	0	0	0
12	Regression coefficient (β)	0	0	0	38	0	0	1

ECO = Ecology, ENC = Environmental Chemistry, ENE = Environmental Engineering, HTM = Health, Toxicology and Mutagenesis, MPL = Management, Monitoring, Policy and Law, NLC = Nature and Landscape Conservation, and WTS = Water and Science Technology.

17
18
19
20
21
22
23
24
25

References

Lamberink, Herm J, Willem M Otte, Michel RT Sinke, Daniël Lakens, Paul P Glasziou, Joeri K Tijdkink, and Christiaan H Vinkers. 2018. “Statistical power of clinical trials increased while effect size remained stable: an empirical analysis of 136,212 clinical trials between 1975 and 2014.” *Journal of Clinical Epidemiology* 102:123–128.

Nord, Camilla L, Vincent Valton, John Wood, and Jonathan P Roiser. 2017. “Power-up: a reanalysis of power failure in neuroscience using mixture modeling.” *Journal of Neuroscience* 37 (34): 8051–8061.

Yang, Yefeng, Alfredo Sánchez-Tójar, Rose E O’Dea, Daniel WA Noble, Julia Koricheva, Michael D Jennions, Timothy H Parker, Malgorzata Lagisz, and Shinichi Nakagawa. 2023. “Publication bias impacts on effect size, statistical power, and magnitude (Type M) and sign (Type S) errors in ecology and evolutionary biology.” *BMC Biology* 21 (1): 71.